

Creation Date 09-Oct-2009

Revision Date 27-Sep-2023

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                                       |
|----------------------------------|-----------------------------------------------------------------------|
| <b>Product Description:</b>      | <b>Phosphorus pentasulfide</b>                                        |
| <b>Cat No. :</b>                 | <b>213640000</b>                                                      |
| <b>Synonyms</b>                  | Sulfur phosphide; Thiophosphoric anhydride; Diphosphorus pentasulfide |
| <b>Index No</b>                  | 015-104-00-1                                                          |
| <b>CAS No</b>                    | 1314-80-3                                                             |
| <b>EC No</b>                     | 215-242-4                                                             |
| <b>Molecular Formula</b>         | P <sub>4</sub> S <sub>10</sub>                                        |
| <b>REACH registration number</b> | -                                                                     |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

|                                                                        |                   |
|------------------------------------------------------------------------|-------------------|
| Flammable solids                                                       | Category 1 (H228) |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 (H260) |

##### Health hazards

|                                             |                   |
|---------------------------------------------|-------------------|
| Acute oral toxicity                         | Category 4 (H302) |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4 (H332) |

##### Environmental hazards

|                        |                   |
|------------------------|-------------------|
| Acute aquatic toxicity | Category 1 (H400) |
|------------------------|-------------------|

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H228 - Flammable solid  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H400 - Very toxic to aquatic life  
H302 + H332 - Harmful if swallowed or if inhaled  
EUH029 - Contact with water liberates toxic gas  
May form combustible dust concentrations in air

#### **Precautionary Statements**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## 2.3. Other hazards

Water reactive  
Lachrymator (substance which increases the flow of tears)  
Stench  
May form explosible dust-air mixture if dispersed  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                 | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                |
|---------------------------|-----------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | 1314-80-3 | EEC No. 215-242-4 | >95      | Flam. Sol. 1 (H228)<br>Water-react. 1 (H260)<br>Acute Tox. 4 (H302)<br>Acute Tox. 4 (H332)<br>Aquatic Acute 1 (H400)<br>(EUH029) |

REACH registration number

-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                     |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                     |
| Skin Contact                       | Immediate medical attention is required. Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                                                                                                                                                                                                                        |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately. Immediate medical attention is required. Remove from exposure, lie down. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person.                                                                                                           |
| Inhalation                         | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. Call a physician or poison control center immediately. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                      |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. May cause pulmonary edema: Symptoms may be delayed

### 4.3. Indication of any immediate medical attention and special treatment needed

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

Contact with water liberates toxic gas.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Contact with water liberates toxic gas. May form explosive mixtures with air. Produce flammable gases on contact with water. Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes. Contact with water liberates toxic gas. Do not allow run-off from fire-fighting to enter drains or water courses. Fine dust dispersed in air may ignite.

#### **Hazardous Combustion Products**

Hydrogen sulfide (H<sub>2</sub>S), Sulfur oxides, Oxides of phosphorus.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Do not expose spill to water.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from open flames, hot surfaces and sources of ignition. Do not taste or swallow. Do not allow contact with water.

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## Hygiene Measures

Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame. Keep in properly labeled containers. Keep from any possible contact with water. Keep away from water or moist air.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 4.3

**Switzerland - Storage of hazardous substances**

Storage class - SC 4.3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                 | European Union                 | The United Kingdom                                                | France                                                      | Belgium                                                                 | Spain                                       |
|---------------------------|--------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------|
| Phosphorus sulfide (P2S5) | TWA: 1 mg/m <sup>3</sup> (8hr) | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr | TWA / VME: 1 mg/m <sup>3</sup> (8 heures). indicative limit | TWA: 1 mg/m <sup>3</sup> 8 uren<br>STEL: 3 mg/m <sup>3</sup> 15 minuten | TWA / VLA-ED: 1 mg/m <sup>3</sup> (8 horas) |

| Component                 | Italy                                                    | Germany                                                       | Portugal                                                                 | The Netherlands                 | Finland                                  |
|---------------------------|----------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------|------------------------------------------|
| Phosphorus sulfide (P2S5) | TWA: 1 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average | TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4 | STEL: 3 mg/m <sup>3</sup> 15 minutos<br>TWA: 1 mg/m <sup>3</sup> 8 horas | TWA: 1 mg/m <sup>3</sup> 8 uren | STEL: 1 mg/m <sup>3</sup> 15 minuutteina |

| Component                 | Austria                                                                            | Denmark                                                                   | Switzerland                                                                | Poland                                                                        | Norway                                                                                      |
|---------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | MAK-KZGW: 2 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 1 mg/m <sup>3</sup> 8 Stunden | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 2 mg/m <sup>3</sup> 15 minutter | STEL: 1 mg/m <sup>3</sup> 15 Minuten<br>TWA: 1 mg/m <sup>3</sup> 8 Stunden | STEL: 3 mg/m <sup>3</sup> 15 minutach<br>TWA: 1 mg/m <sup>3</sup> 8 godzinach | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 3 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component                 | Bulgaria                   | Croatia                                | Ireland                                                            | Cyprus                   | Czech Republic                                                       |
|---------------------------|----------------------------|----------------------------------------|--------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | TWA: 1.0 mg/m <sup>3</sup> | TWA-GVI: 1 mg/m <sup>3</sup> 8 satima. | TWA: 1 mg/m <sup>3</sup> 8 hr.<br>STEL: 3 mg/m <sup>3</sup> 15 min | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 2 mg/m <sup>3</sup> |

| Component                 | Estonia                              | Gibraltar                     | Greece                                                | Hungary                               | Iceland                                                                   |
|---------------------------|--------------------------------------|-------------------------------|-------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | TWA: 1 mg/m <sup>3</sup> 8 tundides. | TWA: 1 mg/m <sup>3</sup> 8 hr | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> 8 órában. AK | TWA: 1 mg/m <sup>3</sup> 8 klukkustundum.<br>Ceiling: 2 mg/m <sup>3</sup> |

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

| Component                 | Latvia                   | Lithuania                     | Luxembourg                         | Malta                    | Romania                        |
|---------------------------|--------------------------|-------------------------------|------------------------------------|--------------------------|--------------------------------|
| Phosphorus sulfide (P2S5) | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> IPRD | TWA: 1 mg/m <sup>3</sup> 8 Stunden | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> 8 ore |

| Component                 | Russia | Slovak Republic          | Slovenia                                                                | Sweden                                    | Turkey                          |
|---------------------------|--------|--------------------------|-------------------------------------------------------------------------|-------------------------------------------|---------------------------------|
| Phosphorus sulfide (P2S5) |        | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> 8 urah<br>STEL: 4 mg/m <sup>3</sup> 15 minutah | TLV: 1 mg/m <sup>3</sup> 8 timmar.<br>NGV | TWA: 1 mg/m <sup>3</sup> 8 saat |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|----------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Phosphorus sulfide (P2S5)<br>1314-80-3 (>95) | DNEL = 2mg/m <sup>3</sup>        |                                     | DNEL = 1mg/m <sup>3</sup>          |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                    | Fresh water        | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|----------------------------------------------|--------------------|----------------------|--------------------|------------------------------------|--------------------|
| Phosphorus sulfide (P2S5)<br>1314-80-3 (>95) | PNEC = 0.00016mg/L |                      |                    |                                    |                    |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                               | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene | See manufacturers recommendations |                 | EN 374      | (minimum requirement) |

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

|     |   |
|-----|---|
| PVC | - |
|-----|---|

## Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure. Impervious clothing. Chemical resistant apron. Boots. Impervious gloves.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                               |                                   |
|-----------------------------------------|-------------------------------|-----------------------------------|
| Physical State                          | Solid                         |                                   |
| Appearance                              | Yellow                        |                                   |
| Odor                                    | Stench                        |                                   |
| Odor Threshold                          | No data available             |                                   |
| Melting Point/Range                     | 286 - 290 °C / 546.8 - 554 °F |                                   |
| Softening Point                         | No data available             |                                   |
| Boiling Point/Range                     | 514 °C / 957.2 °F             | @ 760 mmHg                        |
| Flammability (liquid)                   | Not applicable                | Solid                             |
| Flammability (solid,gas)                | No information available      |                                   |
| Explosion Limits                        | No data available             |                                   |
| Flash Point                             | > 100 °C / > 212 °F           | Method - No information available |
| Autoignition Temperature                | 282 - °C / 539.6 - °F         |                                   |
| Decomposition Temperature               | No data available             |                                   |
| pH                                      | 1                             | 10 g/L (20°C)                     |
| Viscosity                               | Not applicable                | Solid                             |
| Water Solubility                        | reacts forming H3PO4 and H2S  |                                   |
| Solubility in other solvents            | No information available      |                                   |
| Partition Coefficient (n-octanol/water) |                               |                                   |
| Vapor Pressure                          | 1 mmHg @ 300 °C               |                                   |
| Density / Specific Gravity              | 2.080                         |                                   |
| Bulk Density                            | No data available             |                                   |
| Vapor Density                           | Not applicable                | Solid                             |
| Particle characteristics                | No data available             |                                   |

### 9.2. Other information

Molecular Formula P4 S10

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

|                                                                        |                                                                                    |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Molecular Weight                                                       | 444.48                                                                             |
| Flammable solids                                                       | Burning rate or burning time = > 2.2 mm/s or < 45 secs<br>Wetted zone passed - Yes |
| Substances/mixtures which, in contact with water, emit flammable gases | Emitted gas ignites spontaneously Gas(es) = Hydrogen sulfide (H2S)                 |
| Evaporation Rate                                                       | Not applicable - Solid                                                             |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Moisture sensitive. Reacts violently with water.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
Contact with water liberates toxic gas.

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Avoid dust formation. Exposure to moist air or water. Exposure to air or moisture over prolonged periods. Exposure to moisture.

### 10.5. Incompatible materials

Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Hydrogen sulfide (H2S). Sulfur oxides. Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

| Component                 | LD50 Oral                              | LD50 Dermal           | LC50 Inhalation |
|---------------------------|----------------------------------------|-----------------------|-----------------|
| Phosphorus sulfide (P2S5) | 791 mg/kg ( Rat )<br>389 mg/kg ( Rat ) | 3160 mg/kg ( Rabbit ) | -               |

#### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

#### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

|                             |                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| (e) germ cell mutagenicity; | Based on available data, the classification criteria are not met                                                              |
| (f) carcinogenicity;        | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | Based on available data, the classification criteria are not met                                                              |
| (h) STOT-single exposure;   | Based on available data, the classification criteria are not met                                                              |
| (i) STOT-repeated exposure; | Based on available data, the classification criteria are not met                                                              |
| Target Organs               | None known.                                                                                                                   |
| (j) aspiration hazard;      | Not applicable<br>Solid                                                                                                       |

**Symptoms / effects, both acute and delayed** May cause pulmonary edema. Symptoms may be delayed.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

**Ecotoxicity effects** Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component                 | Freshwater Fish | Water Flea             | Freshwater Algae |
|---------------------------|-----------------|------------------------|------------------|
| Phosphorus sulfide (P2S5) |                 | EC50 = 0.16 mg/L (48h) |                  |

### 12.2. Persistence and degradability

**Persistence** Not readily biodegradable  
**Degradability** Persistence is unlikely, based on information available, Soluble in water.  
**Degradation in sewage treatment plant** Not relevant for inorganic substances, Reacts with water.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Product does not bioaccumulate due to reaction with water; Bioaccumulation is unlikely

### 12.4. Mobility in soil

Reacts with water The product is water soluble, and may spread in water systems Is not likely mobile in the environment. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

Water reactive.

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment. Do not empty into drains.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN1340

#### 14.2. UN proper shipping name

PHOSPHORUS PENTASULPHIDE

#### 14.3. Transport hazard class(es)

4.3

#### Subsidiary Hazard Class

4.1

#### 14.4. Packing group

II

### ADR

#### 14.1. UN number

UN1340

#### 14.2. UN proper shipping name

PHOSPHORUS PENTASULPHIDE

#### 14.3. Transport hazard class(es)

4.3

#### Subsidiary Hazard Class

4.1

#### 14.4. Packing group

II

### IATA

#### 14.1. UN number

UN1340

#### 14.2. UN proper shipping name

PHOSPHORUS PENTASULPHIDE

#### 14.3. Transport hazard class(es)

4.3

#### Subsidiary Hazard Class

4.1

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## 14.4. Packing group II

**14.5. Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                 | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Phosphorus sulfide (P2S5) | 1314-80-3 | 215-242-4 | -      | -   | X     | X    | KE-12128 | X    | X    |

| Component                 | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Phosphorus sulfide (P2S5) | 1314-80-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH Not applicable

| Component                 | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | 1314-80-3 | -                                                                   | -                                                                             | -                                                                                                     |

#### Seveso III Directive (2012/18/EC)

| Component                 | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Phosphorus sulfide (P2S5) | 1314-80-3 | Not applicable                                                                            | Not applicable                                                                           |

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component                 | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------------------|---------------------------------------|-------------------------|
| Phosphorus sulfide (P2S5) | WGK3                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H228 - Flammable solid

H260 - In contact with water releases flammable gases which may ignite spontaneously

H302 - Harmful if swallowed

H332 - Harmful if inhaled

H400 - Very toxic to aquatic life

EUH029 - Contact with water liberates toxic gas

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Training Advice

Chemical incident response training.

# SAFETY DATA SHEET

Phosphorus pentasulfide

Revision Date 27-Sep-2023

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|                  |                 |
|------------------|-----------------|
| Creation Date    | 09-Oct-2009     |
| Revision Date    | 27-Sep-2023     |
| Revision Summary | Not applicable. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**